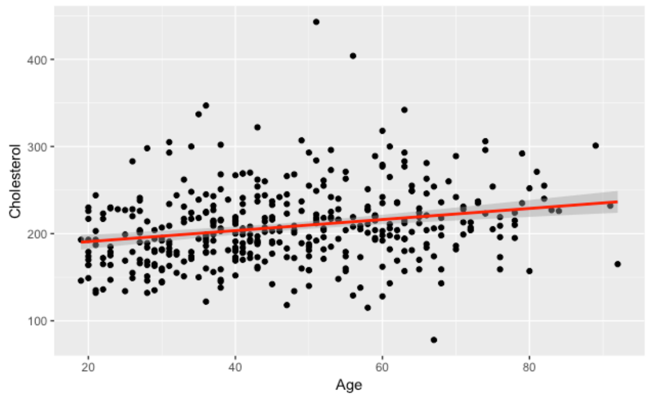
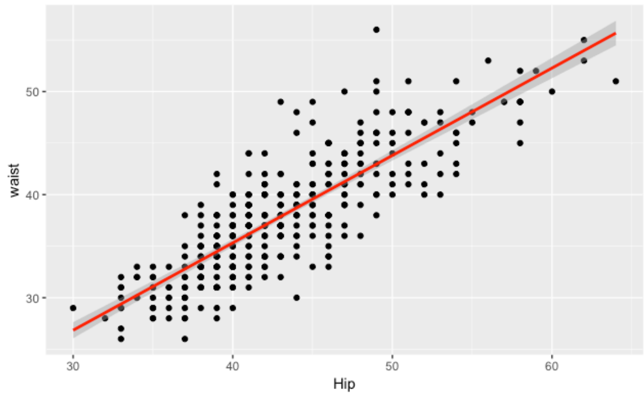
9 Regression analysis

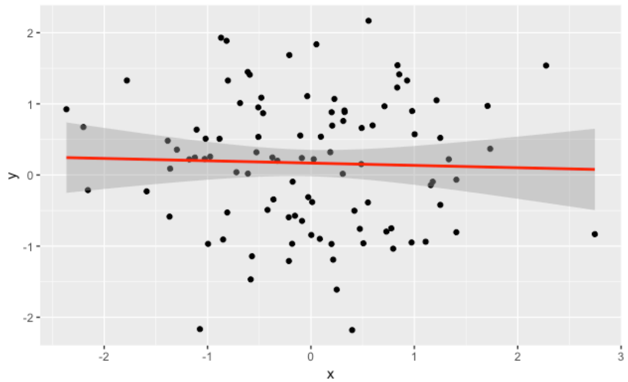
Predicting variables from other variables

Relation between numerical variables

- How easy is it to draw a line through a scatter plot?







Relation between numerical variables

- Variance: $$Var(x) = (s_x)^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$$ dimension: $[x]^2$

- Covariance : $$Cov(x, y) = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})$$ dimension: [x][y]

- Pearson Correlation : $$Corr(x, y) = r = \frac{1}{N-1}\sum_{i=1}^{N}\frac{(x_i - \bar{x})}{s_x}\frac{(y_i - \bar{y})}{s_y}$$ dimension: none

- Properties:
  - correlation is scale invariant, covariance is not!
  - cor(x,x) = 1
  - -1 =< cor(x,y) =< +1

Relation between numerical variables

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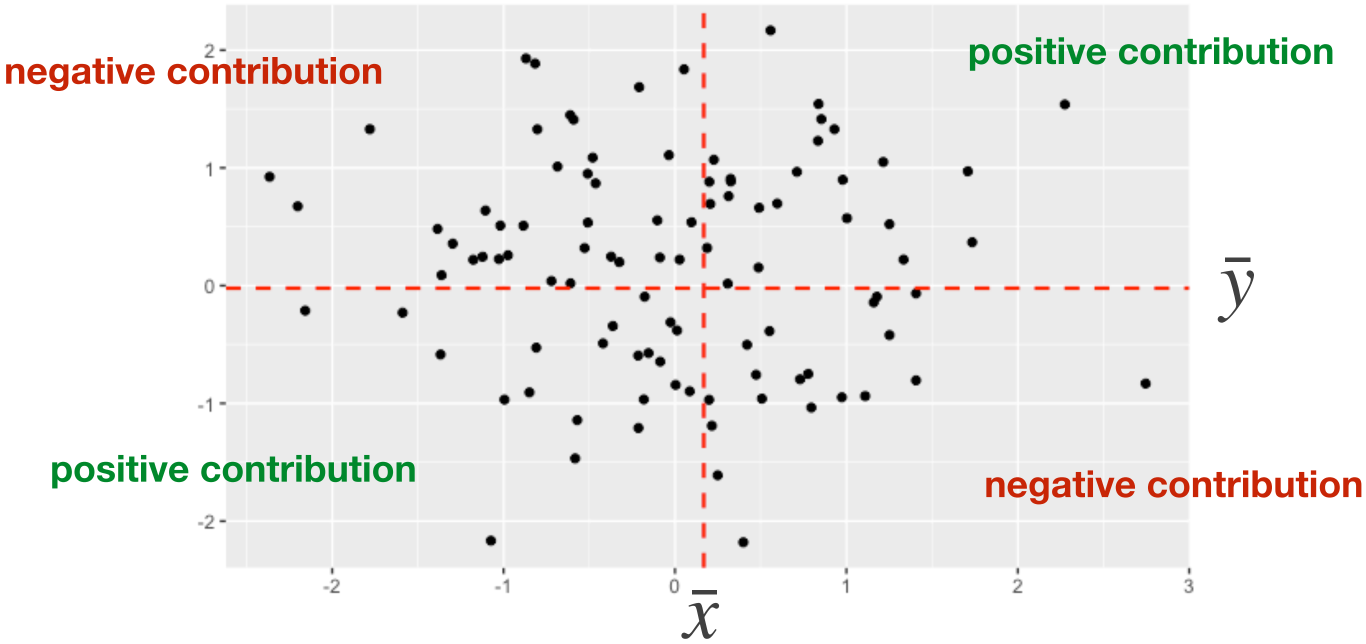
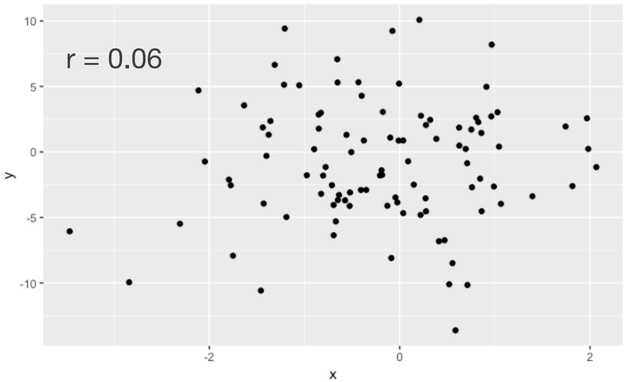
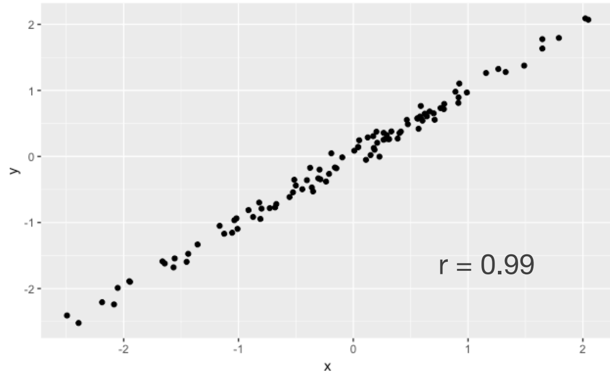


$$Corr(x,y) = \frac{1}{N-1} \sum_{i=1}^{N} \frac{(x_i - \bar{x})}{s_x} \frac{(y_i - \bar{y})}{s_y}$$

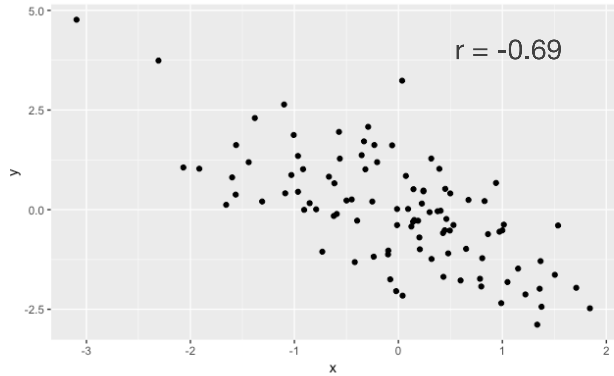
Relation between numerical variables





These are sample-based estimations of the correlation

→ what about the population correlation?



Statistical test on correlation

- the sample correlation coefficient ***r*** is an estimate of the real unknown correlation coefficient **ρ**
- Hypothesis test: ***could ρ actually be zero?***
- t-test with H_0: $\rho = 0$

$$t = \frac{r}{se_r}$$

estimate

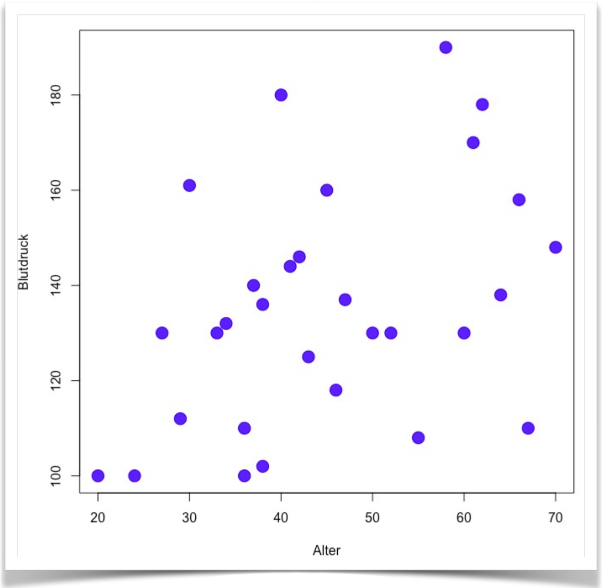
standard error

$$se_r = \sqrt{\frac{1 - r^2}{n - 2}}$$

- H_0 distribution: t-distribution with n-2 degrees of freedom

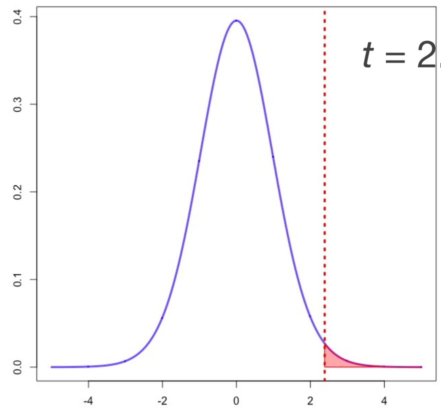
Example

n = 30




```
> cor.test(diab[1:30,7],diab[1:30,12])

    Pearson's product-moment correlation

data:  diab[1:30, 7] and diab[1:30, 12]
t = 2.386, df = 28, p-value = 0.02404
alternative hypothesis: true correlation is
not equal to 0
95 percent confidence interval:
 0.05960801 0.67182894
sample estimates:
    cor
0.41105
```

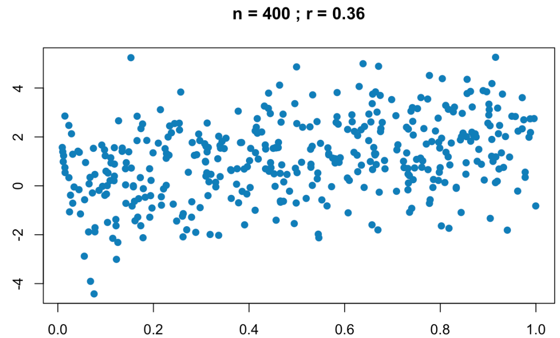
$$t = \frac{r}{se_r} \qquad se_r = \sqrt{\frac{1 - r^2}{n - 2}}$$



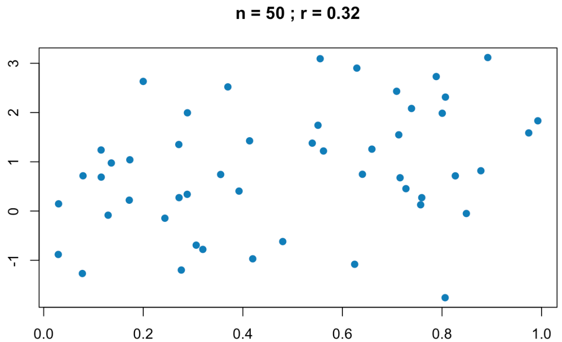
$t = 2.386$

$p_{1sided} = 0.012$
$p_{2sided} = 0.024$

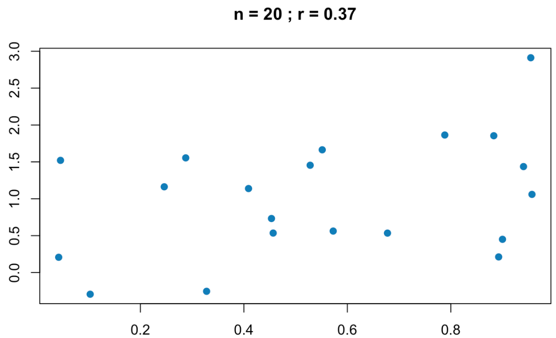
Significance of correlation




```
         Pearson's product-moment correlation

data:  x and y
t = 7.6555, df = 398, p-value = 1.473e-13
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.2696858 0.4408299
sample estimates:
      cor
0.3582639
```



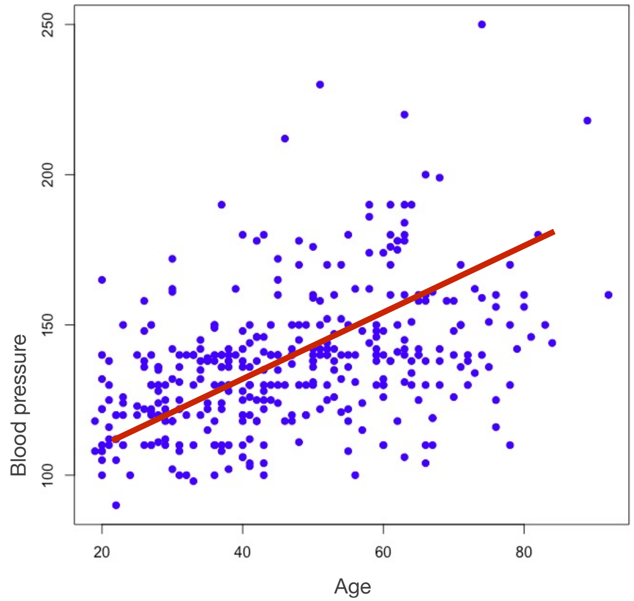

```
         Pearson's product-moment correlation

data:  x and y
t = 2.3721, df = 48, p-value = 0.02175
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.05008513 0.55245463
sample estimates:
      cor
0.3239171
```




```
         Pearson's product-moment correlation

data:  x and y
t = 1.7031, df = 18, p-value = 0.1058
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.08380564  0.69970356
sample estimates:
      cor
0.3725307
```

Regression models

- **Predict one numerical variable using one or several other numerical variables**
- Different questions:
  - *is there a relation between age and blood pressure?* → **Correlation**
  - *how well can I predict blood pressure from age?* → **Regression model**
- Principle
  - **learn** regression model from data (*training*)
  - **test** validity on independent datasets (*testing*)
  - **predict** on new data (*predict*)



Linear regression

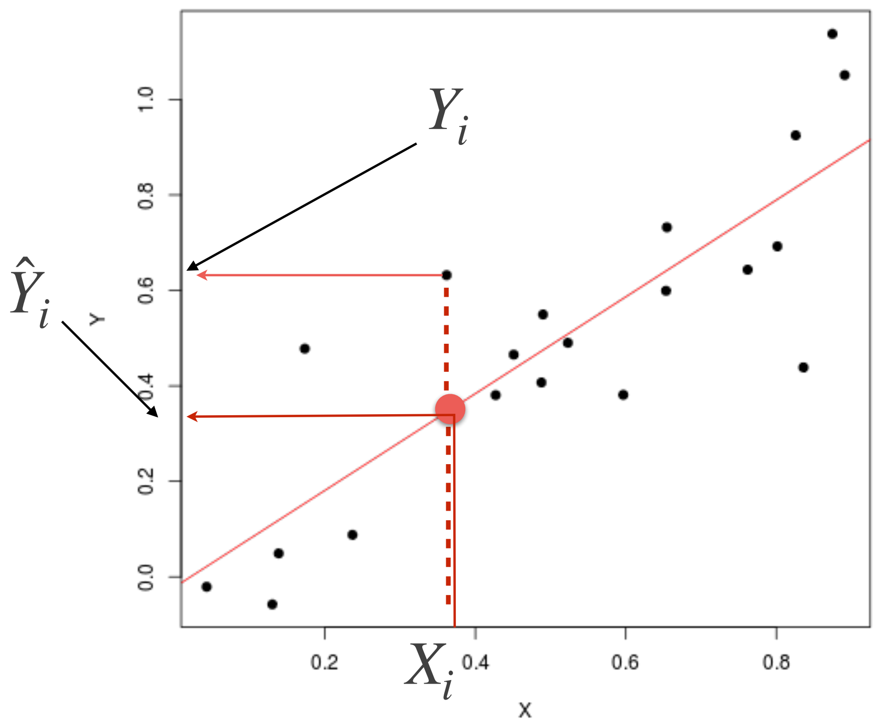
- We assume a **linear relationship** between variables (X,Y)

$$X = \{X_1, X_2, \ldots, X_n\} \quad Y = \{Y_1, Y_2, \ldots, Y_n\}$$

$$\hat{Y}_i = b_0 + b_1 X_i$$

- For each X_i, we can estimate a value $\hat{Y}_i$

- b_0 = intercept
  b_1 = slope

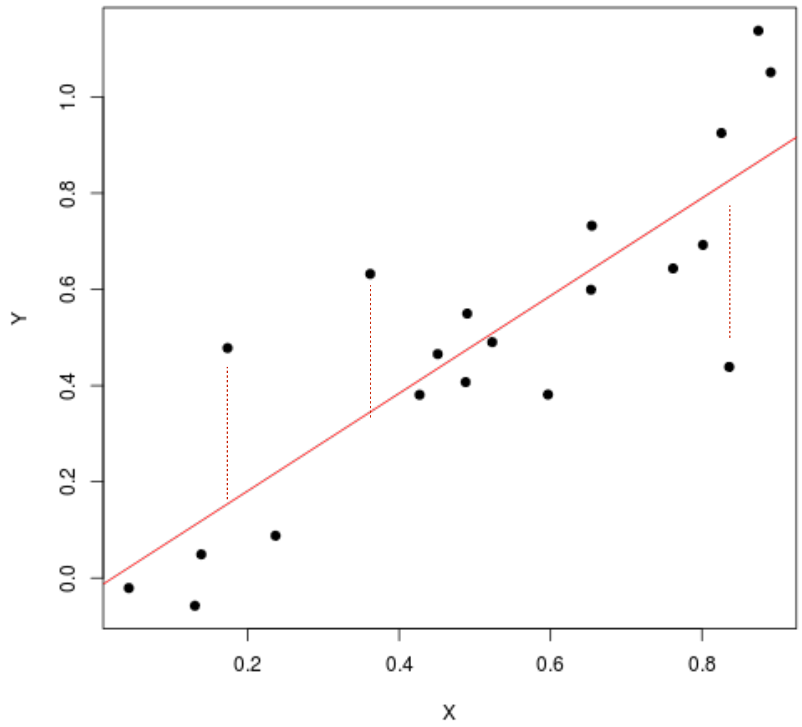


Least square

- Parameters of the regression line are estimated using **least-square method**
  → minimize the sum of squares of the deviations

$$\min_{b_0, b_1} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

$$\boxed{\begin{aligned} b_0 &= \bar{Y} - b_1 \cdot \bar{X} \\ b_1 &= corr(X, Y)\frac{s_Y}{s_X} \end{aligned}}$$



Residuals

- Estimated value and real value do not generally coincide

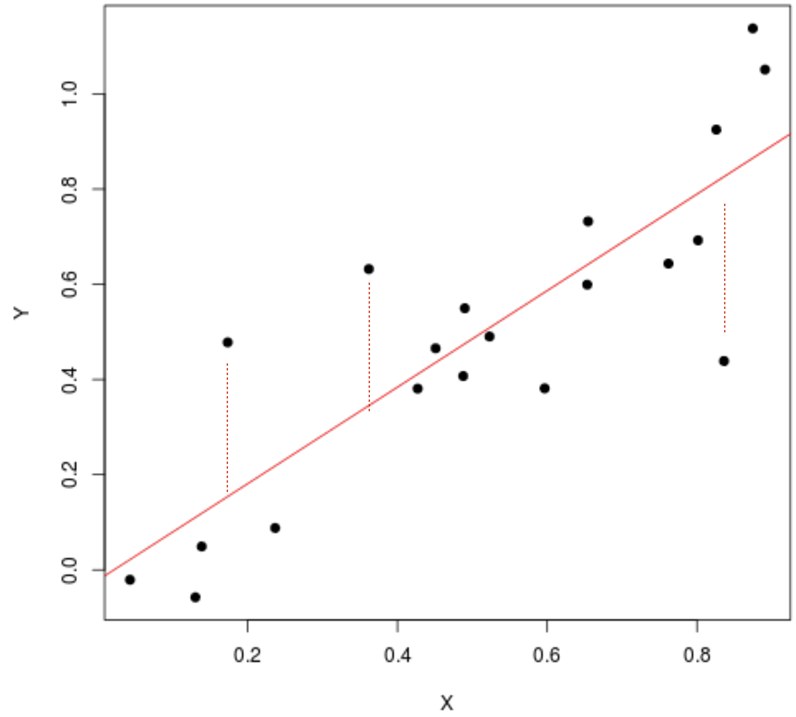
$$\hat{Y}_i \neq Y_i$$

- we have

$$Y_i = b_0 + b_1 X_i + e_i = \hat{Y}_i + e_i$$

- the e_i are called **residuals**

$$e_i = Y_i - \hat{Y}_i$$



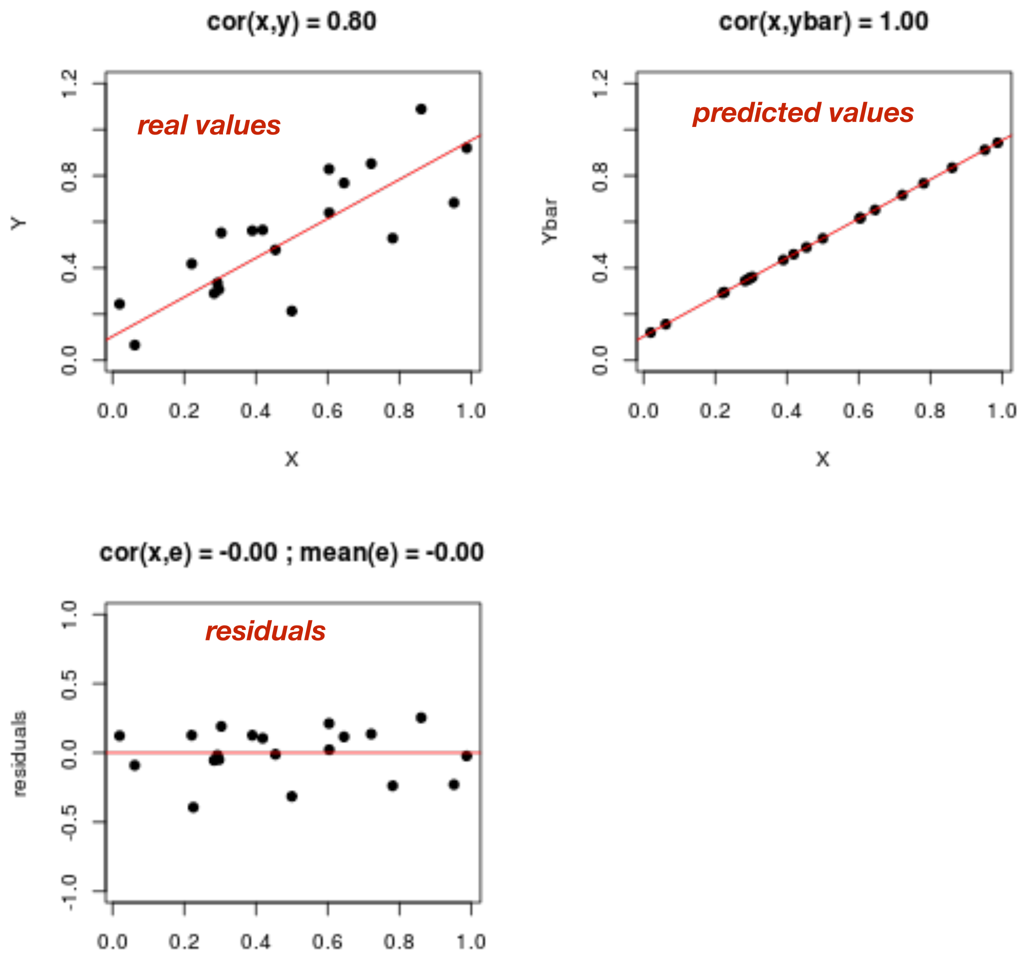
Residuals

- all the influence of X has been "absorbed" by $\hat{Y}$

- X should have no influence on the residuals e_i

$$corr(X, \hat{Y}) = 1$$

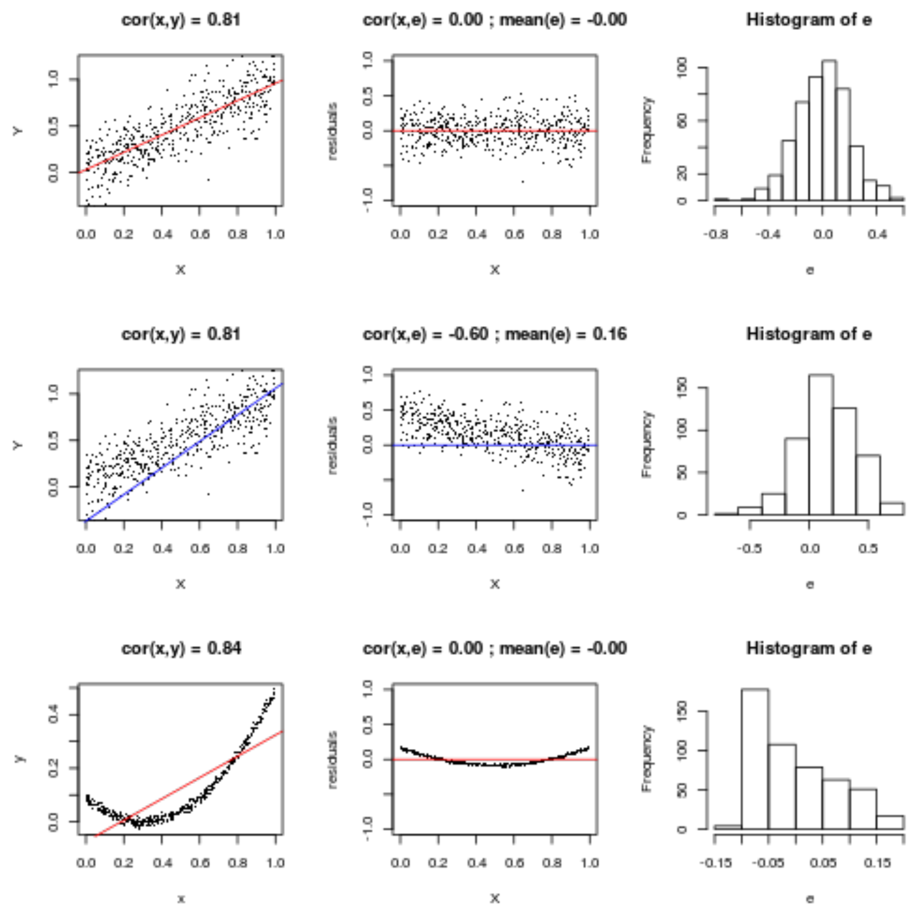
$$corr(X, e) = 0$$



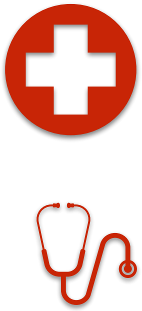
Residuals

- residuals should
  - not correlate with X
  - have mean value 0
  - be normally distributed

- If this is not the case, the linearity assumption is not true!
- Important quality assessment!

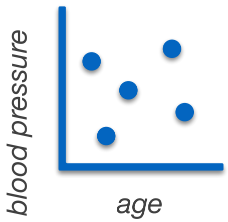


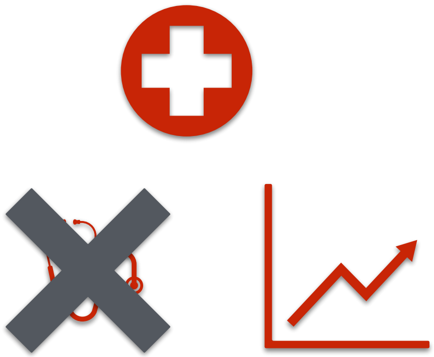
Evaluating a regression model



Device to measure BP

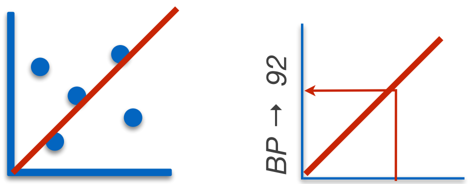
1. Ask for age
2. Measure blood pressure

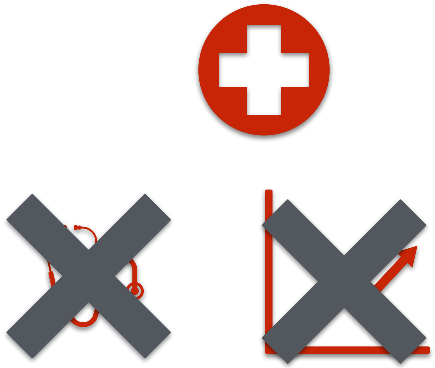




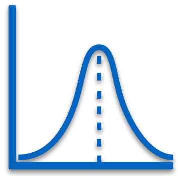
Stats knowledge

1. Ask for age
2. Build regression model from last year's patients
3. Predict BP





1. Ask for age
2. Compute average BP of all patients last year
3. Predict BP using average

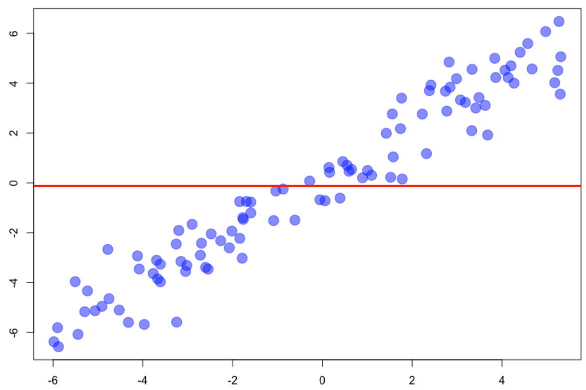


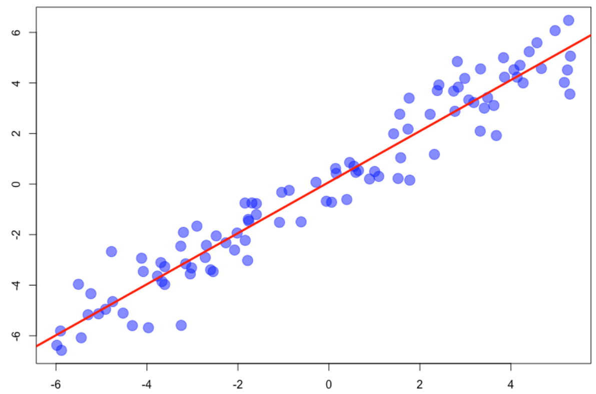
BP = 87

Evaluating a regression model

- Does the regression model work better than the simple null-model?
- Probably more accurate, but also more complex (more parameters)
- Is this improvement worth the higher complexity?

Null-model:
blood pressure = mean
blood pressure over all patients





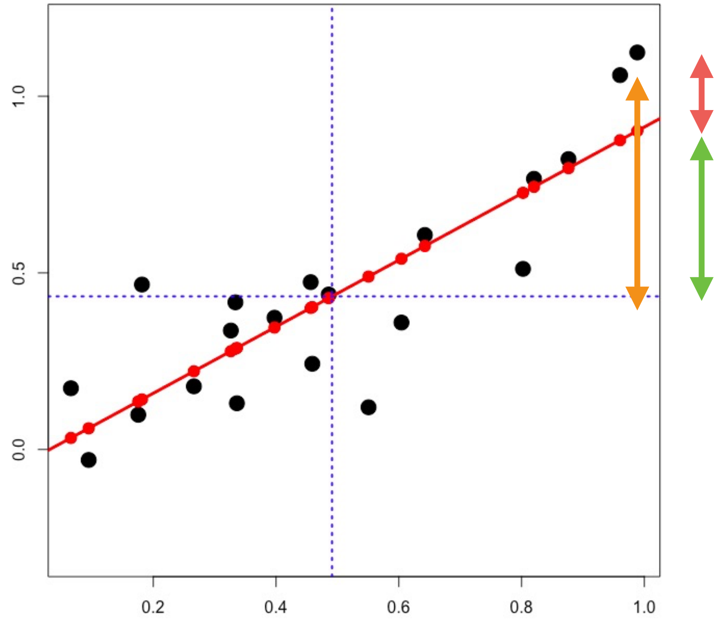
Regression-model:
blood pressure = b0 + b1 Age

We need to evaluate how much of the variance of the data is explained by the model

Variance decomposition

- Variance of Y can be decomposed in different components
  - variance of $\hat{Y}$
  - variance of the residuals e
- because $corr(\hat{Y}, e) = 0$

$$Var(Y) = Var(\hat{Y} + e) = Var(\hat{Y}) + Var(e)$$



$$\underbrace{\boxed{\sum_{i=1}^{n} (Y_i - \bar{Y})^2}}_{\text{Total sum of squares}}^{SS_T} = \underbrace{\boxed{\sum_{i=1}^{n} (\hat{Y}_i - \bar{Y})^2}}_{\text{Model sum of squares}}^{SS_M} + \underbrace{\boxed{\sum_{i=1}^{n} e_i^2}}_{\text{Residuals sum of squares}}^{SS_R}$$

Variance decomposition

$$\underbrace{\boxed{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}_{\text{Total sum of squares}}^{SS_T} = \underbrace{\boxed{\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2}}_{\text{Model sum of squares}}^{SS_M} + \underbrace{\boxed{\sum_{i=1}^{n} e_i^2}}_{\text{residuals sum of squares}}^{SS_R}$$

- if the fit of the regression model is good, SS_R should be small and SS_M large

$$R^2 = \frac{SS_M}{SS_T} \in [0,1]$$

- R^2 is the **proportion of variance explained by the model**
- we have $\text{corr}(X,Y)^2 = R^2$
- example: corr(X,Y) = 0.6 : the linear regression model explains 36% of the total variance

Variance decomposition

$$\underbrace{\boxed{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}_{\text{Total sum of squares}}^{SS_T} = \underbrace{\boxed{\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2}}_{\text{Model sum of squares}}^{SS_M} + \underbrace{\boxed{\sum_{i=1}^{n} e_i^2}}_{\text{residuals sum of squares}}^{SS_R}$$

- explained variance $SS_M = SS_T \cdot R^2$
- variance not explained $SS_R = SS_T \cdot (1 - R^2)$
- Ratio

$$F = \frac{\bar{SS}_M}{\bar{SS}_R} \qquad \bar{SS}_M = SS_M \qquad df = 1$$

number of explanatory variables

$$\bar{SS}_R = \frac{1}{n-2} SS_R \qquad df = n - 2$$

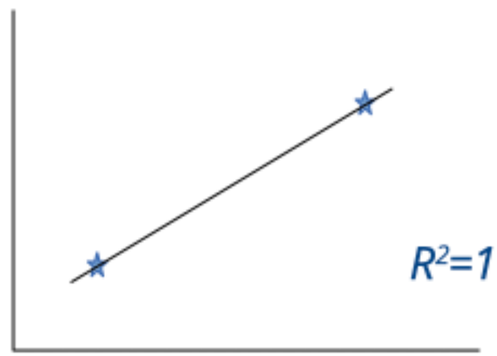
number of coefficients to estimate
= expl. variables +1

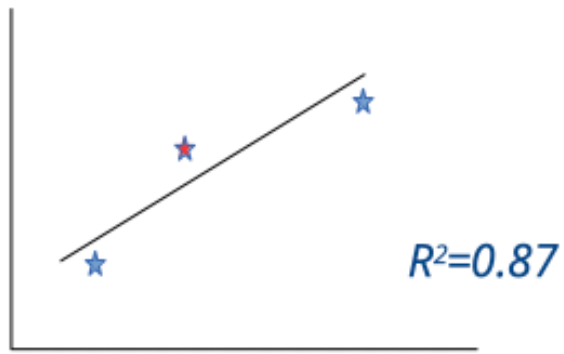
- The larger *F* the better the model describes the data

Degrees of freedom in regression models?

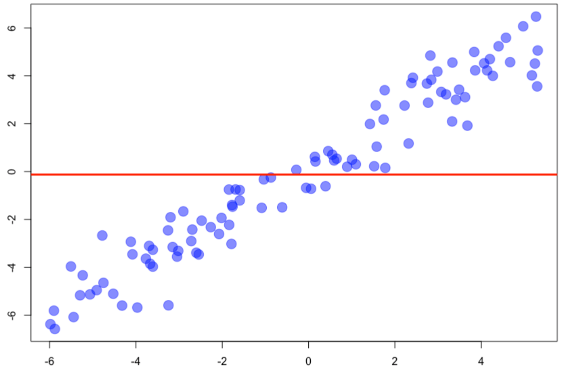
- How many degrees of freedom do we have in a simple linear regression model?
- or: ***how many data points do I need to perform a regression analysis?***
  - ***n = 1*** : single point → any line can be drawn through this point
  - ***n = 2*** : there is a single line going through both points → $R^2 = 1$
    This line does not describe a general relationship between x and y
    → 0 degrees of freedom
  - ***n = 3*** : we can build a regression model with $R^2 < 1$
    → 1 degree of freedom

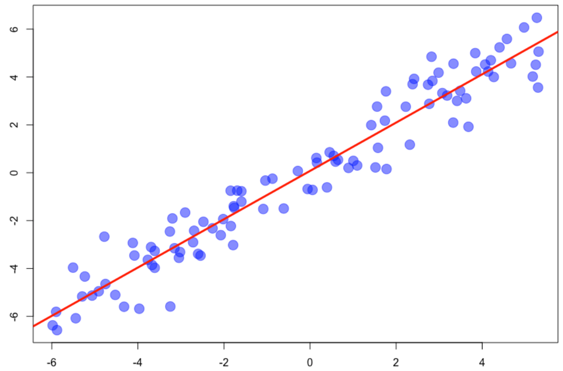
    df = n - 2





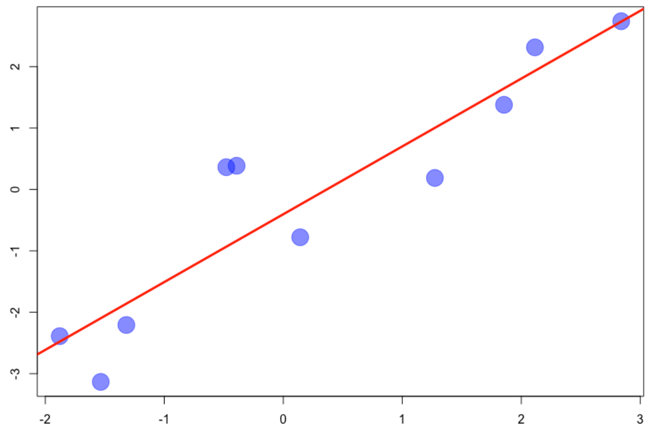
F-test

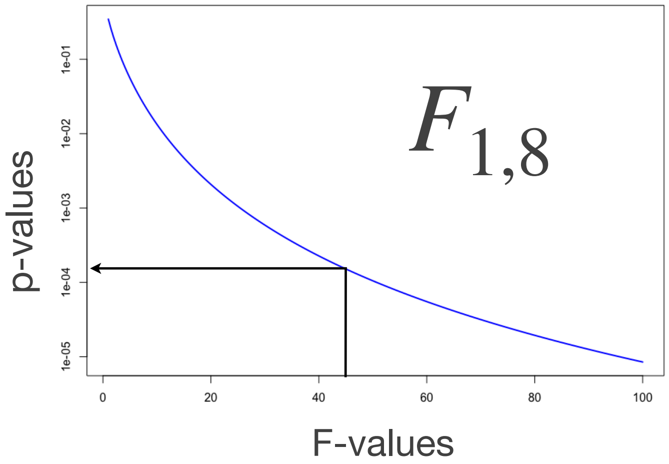




- With the F-test, we can compare 2 models
  - **null model** : $Y_i = b_0$ with b_0 = mean(Y)
  - **full model:** $Y_i = b_0 + b_1 X_i$
- Null hypothesis:
  - full-model not significantly better than null-model
  - or: $b_1 = 0$
  - under H_0 : $F \sim F_{1,n-2}$

Example of regression

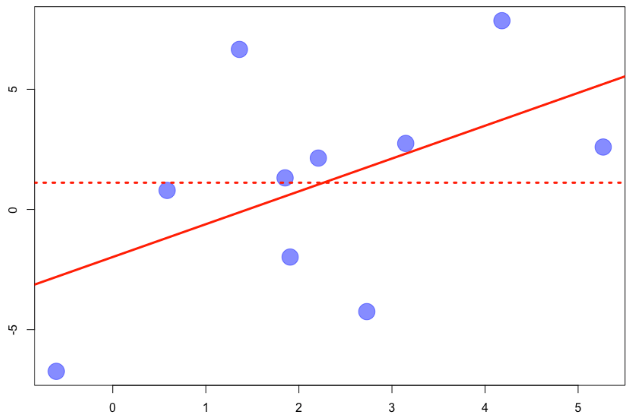


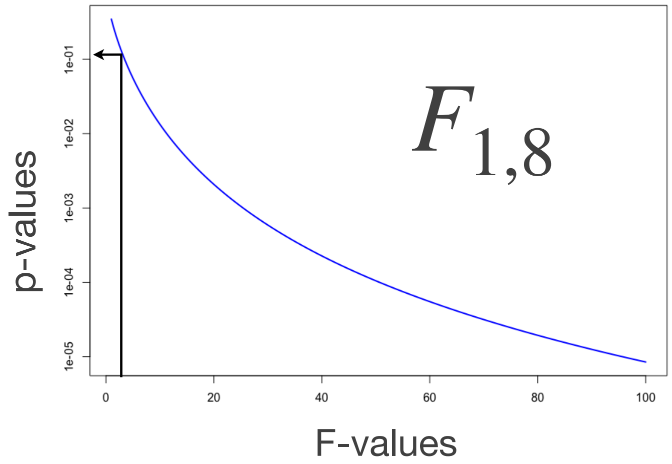


- 10 data points F = 43.93

- Degrees of freedom
  - Residuals : df = 1
  - Full model: df = 8
  - p = 0.00016

Example of regression





- 10 data points F = 2.88

- Degrees of freedom
  - Residuals : df = 1
  - Full model: df = 8
  - **p = 0.12**

y-values could be as well predicted using the mean of y

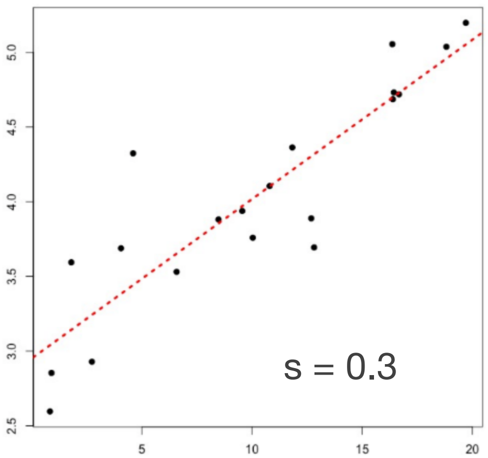
Testing coefficients

- if $b_1 = 0$, then Y cannot be predicted using X
- Reverse statement: if b_1 is significantly different from 0, then X can help predict Y
- Beware
  - a small b_1 value can significantly be different from 0
  - a large b_1 value can be compatible with $b_1=0$
- Deviation from b1 = 0 can be tested using a **t-test**

standard deviation of the residues

$$t = \frac{b_1}{se_{b_1}} \qquad se_{b_1} = \frac{s}{\sqrt{\sum_{i=1}^{n} (X_i - \bar{X})^2}}$$

Hypothesis testing for regression coefficients

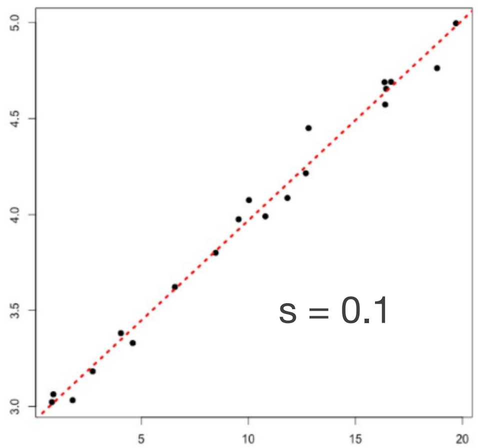


b0

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.95159    0.15391  19.177 1.99e-13 ***
x            0.10668    0.01309   8.147 1.89e-07 ***
```

b1

$$y = 0.1x + 3$$

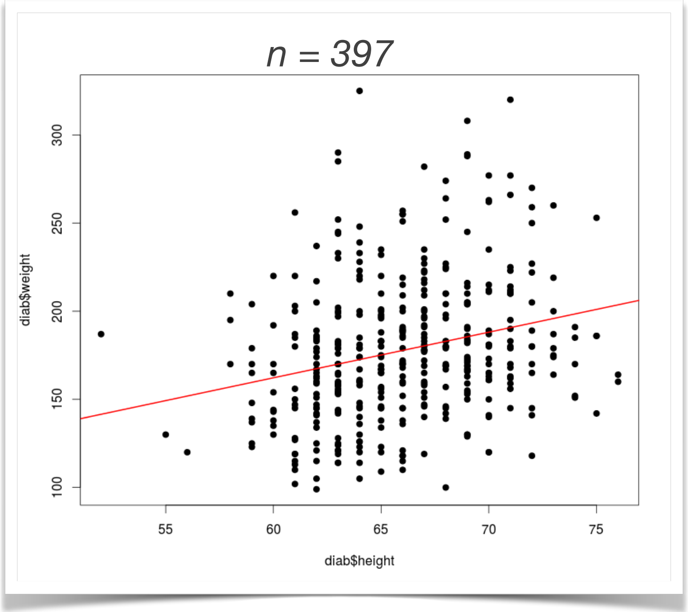



```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.926370   0.032410   90.29   <2e-16 ***
x           0.104333   0.002757   37.84   <2e-16 ***
```

large spread in the data can lead to high uncertainty in regression coefficients

Example of linear regression

height → weight ?

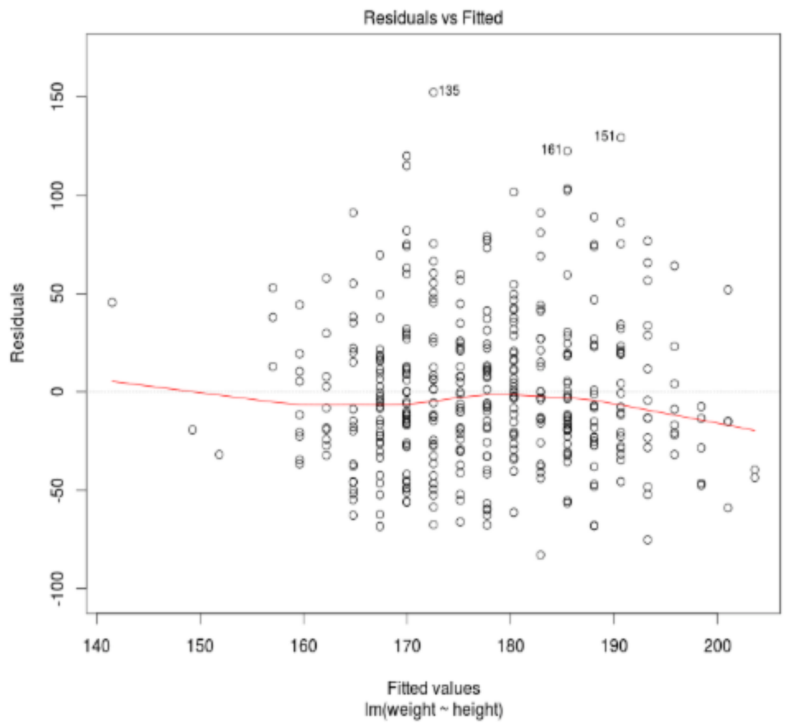



```
> l <- lm(weight ~ height,data=diab)
> summary(l)
Call:
lm(formula = weight ~ height, data = diab)
Residuals:
    Min      1Q  Median      3Q     Max
-82.906 -26.380  -6.731  21.331 152.445
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.9422    33.1694   0.209    0.834
height        2.5877     0.5016   5.159 3.94e-07 ***
---
Residual standard error: 39.13 on 395 degrees of freedom
  (6 observations deleted due to missingness)
Multiple R-squared:  0.06313,   Adjusted R-squared:  0.06076
F-statistic: 26.62 on 1 and 395 DF,  p-value: 3.938e-07
```

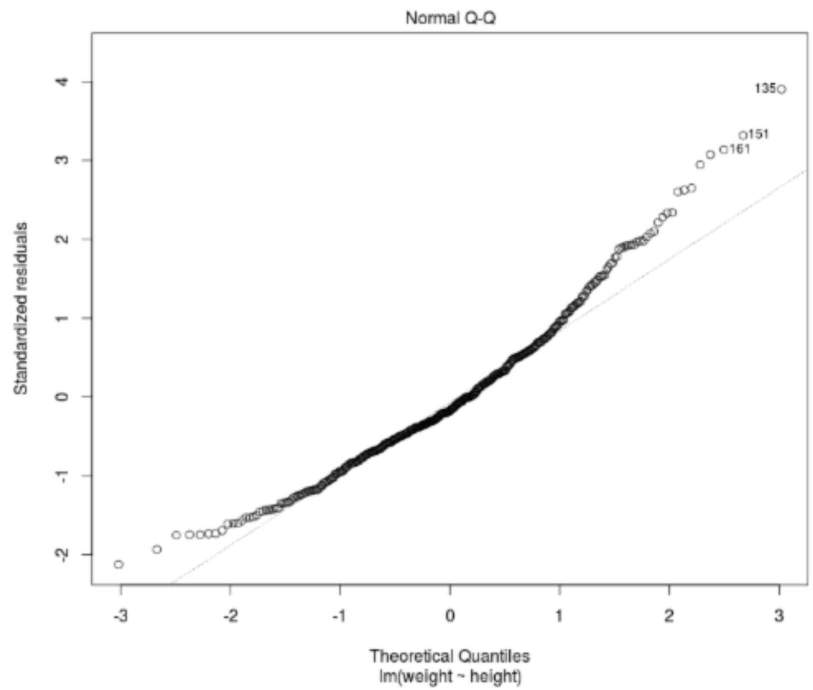
What can we learn?

- only 6% of the variance can be explained by the regression model (R^2)
- b_1 coefficient is significantly different from 0 (t-test p-value)
- Regression model is significantly better than null-model (F-test p-value)

Diagnostic plots



Residuals are independent of X variable

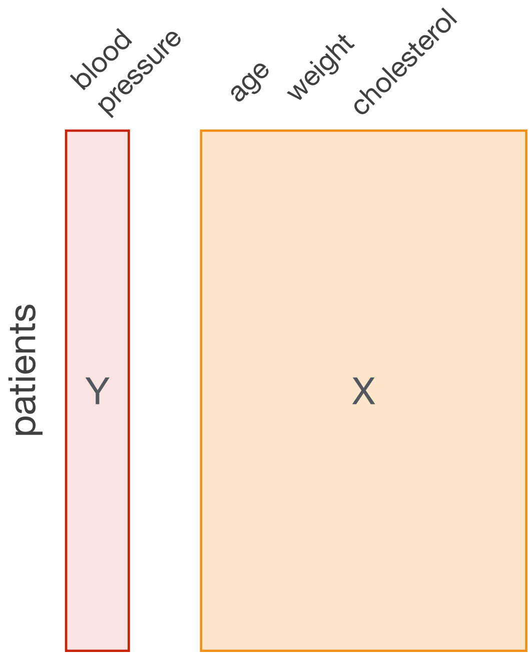


Residuals are (kind of) normally distributed

Multiple Regression

$$Y_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + \cdots + b_r X_{ri} + e_i$$

- Y is the variable to be explained (e.g. blood pressure)
- i = 1,…, n are the **observations** (e.g. patients)
- k = 1,…, r are the **explanatory variables** (e.g. age, cholesterol, weight, …)



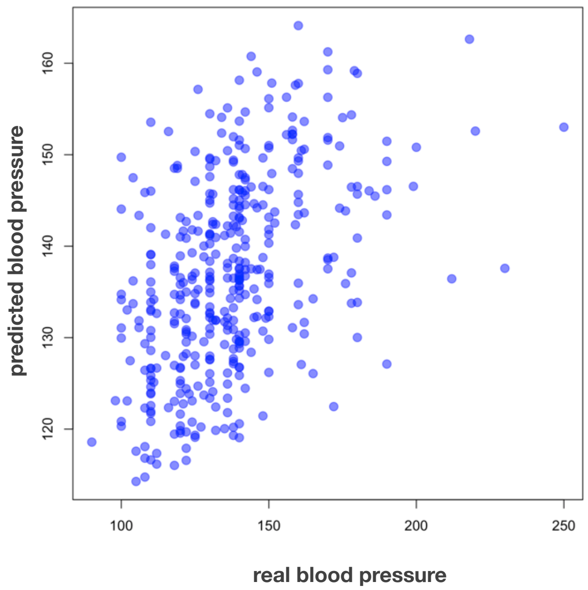
Example

- **blood pressure ~ age + weight + cholesterol**

```
Call:
lm(formula = bp.1s ~ age + weight + chol, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-49.725 -12.786  -1.705   9.603  96.990

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 88.22918    6.78992  12.994   <2e-16 ***
age          0.59525    0.06391   9.314   <2e-16 ***
weight       0.06093    0.02536   2.403   0.0167 *
chol         0.04789    0.02365   2.025   0.0435 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.2 on 392 degrees of freedom
  (7 observations deleted due to missingness)
Multiple R-squared:  0.2192,	Adjusted R-squared:  0.2132
F-statistic: 36.68 on 3 and 392 DF,  p-value: < 2.2e-16
```

t-test

F-test



t-test on regression coefficients

$$BP_i = b_0 + b_1 \text{ age}_i + b_2 \text{ weight}_i + b_3 \text{ chol}_i + e_i$$

Could this coefficient be equal to 0?

or: Does weight contribute to explain blood pressure?

Standard deviation of residuals

$$t = \frac{b_2}{se_{b_2}} \qquad se_{b_2} = \frac{s}{\sqrt{(1 - \boxed{R^2_{X_2, X_{l \neq 2}}})\ \boxed{s^2_{X_2}}\ (n-1))}}$$

R^2 of the regression of weight with all other variables

Standard deviation of variable weight

Beware of correlated variables!

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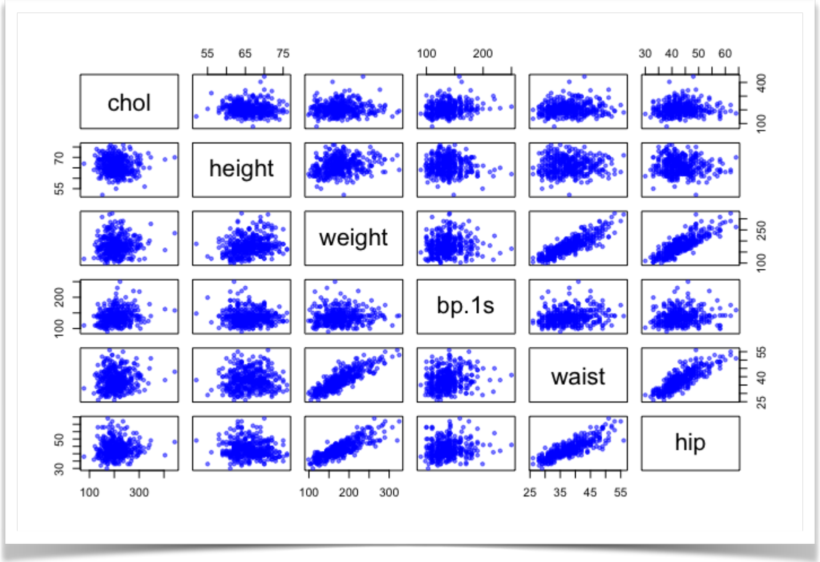
UNIVERSITÄT HEIDELBERG
ZUKUNFT SEIT 1386

$$t = \frac{b_2}{se_{b_2}} \qquad se_{b_2} = \frac{s}{\sqrt{\left(1 - \boxed{R^2_{X_2, X_{l \neq 2}}}\right) s^2_{X_2} \, (n-1))}}$$

- If weight (= X_2) is **strongly correlated with another variable** (=X_l), then R^2 ~1
- Hence, the **standard error *se* become very large**
- The t coefficient become very small
- Test is no longer significant

Beware of correlated variables

- **Highly correlated variables should be avoided in a regression model!**
- Possible solutions:
  - inspect **pairwise scatter-plots / correlations** and eliminate variables if strongly correlated to others
  - perform **principal component analysis** on the explanatory variables, and use the PCs as explanatory variables (Remember: PCs are NOT correlated with each other)



F-statistics

$$\underbrace{\boxed{\sum_{i=1}^{n}(Y_i-\bar{Y})^2}}_{\text{Total sum of squares}}^{SS_T}=\underbrace{\boxed{\sum_{i=1}^{n}(\hat{Y}_i-\bar{Y})^2}}_{\text{Model sum of squares}}^{SS_M}+\underbrace{\boxed{\sum_{i=1}^{n}e_i^2}}_{\text{residuals sum of squares}}^{SS_R}$$

number of explanatory variables

$$F=\frac{\bar{SS}_M}{\bar{SS}_R}$$

$$\bar{SS}_M=\frac{1}{r}SS_M \qquad df=r$$

$$\bar{SS}_R=\frac{1}{n-(r+1)}SS_R \qquad df=n-(r+1)$$

number of coefficients to estimate = expl. variables +1

H0 hypothesis: full model is not better than model with $Y = b_0$

$$H_0:\ F\sim F_{r,n-(r+1)}$$

F-test

- **blood pressure ~ age + weight + cholesterol**

```
Call:
lm(formula = bp.1s ~ age + weight + chol, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-49.725 -12.786  -1.705   9.603  96.990

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 88.22918    6.78992  12.994  <2e-16 ***
age          0.59525    0.06391   9.314  <2e-16 ***
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```
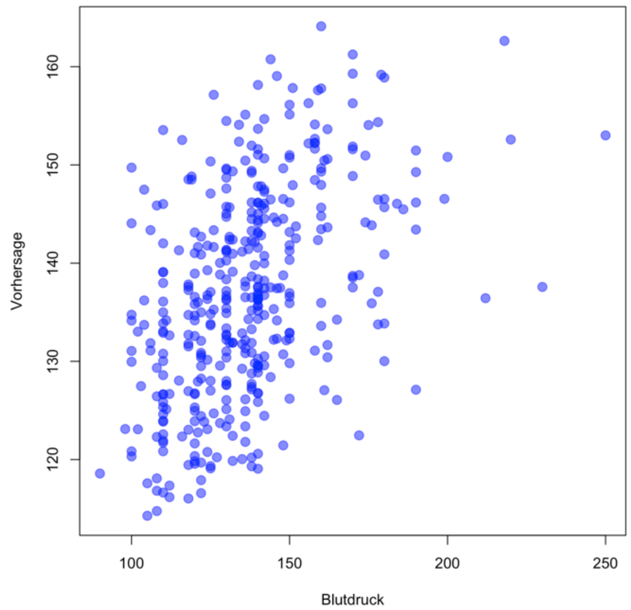

Correlation predicted values / real values
$R^2 = 0.2192$



The F-test tests if ALL coefficients could be zero all together

Comparing models

- Does the inclusion of additional explanatory variables necessarily improve the model?
- **Model 1:** 1 variable

$$BP_i = b_0 + b_1 \text{age}_i + e_i \qquad SS_R^1 = \sum_{i=1}^{n} e_i e_i$$

- **Model 2:** 3 variables

$$BP_i = \tilde{b}_0 + \tilde{b}_1 \text{ age}_i + \tilde{b}_2 \text{ weight}_i + \tilde{b}_3 \text{ chol}_i + \tilde{e}_i \qquad SS_R^2 = \sum_{i=1}^{n} \tilde{e}_i \tilde{e}_i$$

- Is model 2 significantly better than model 1?

$$F = \frac{SS_R^1 - SS_R^2}{3-1} / \frac{SS_R^2}{n-(3+1)} \qquad H_0 : \quad F \sim F_{3-1,n-3-1}$$

Comparing models

$$BP_i = b_0 + b_1 \text{age}_i + e_i$$

```
Call:
lm(formula = bp.1s ~ age, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-49.985 -12.796  -1.836   9.309  96.313 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 108.03563    3.11026  34.735   <2e-16 ***
age           0.61691    0.06258   9.857   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 
1

Residual standard error: 20.42 on 394 degrees of freedom
Multiple R-squared:  0.1978,  Adjusted R-squared:  0.1958 
F-statistic: 97.17 on 1 and 394 DF,  p-value: < 2.2e-16
```

$$BP_i = \tilde{b}_0 + \tilde{b}_1 \text{ age}_i + \tilde{b}_2 \text{ weight}_i + \tilde{b}_3 \text{ chol}_i + \tilde{e}_i$$

```
Call:
lm(formula = bp.1s ~ age + weight + chol, data = data)

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```

Both models are better than the null-model
(check the F-test)
But is model 2 better than model 1?

Comparing models

$$BP_i = b_0 + b_1 \text{age}_i + e_i$$

```
Call:
lm(formula = bp.1s ~ age, data = data)

Residuals:
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```

$$F = \frac{SS_R^1 - SS_R^2}{3-1} \Big/ \frac{SS_R^2}{n-(3+1)}$$

	degrees of freedom	SS_R	F	P-value
Model 1	396-2	164349		
Model 2	396-4	159978	5,3559	0,005072

Model 2 represents a significant improvement w.r.t. Model 1 !

Week 8 - stop and think

First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!

- We consider the following regression model to describe the growth of a plant, depending on the amount of Sunlight received (time, hours) and Water (ml)
- Model A: **Height = 5.2 + 3.8 × Sunlight**
  - Coefficient for Sunlight: $b_1 = 3.8$, SE = 0.6, t = 6.33, $p < 0.001$
  - $R^2 = 0.45$
- Model B: **Height = 2.1 + 3.2 × Sunlight + 0.8 × Water**
  - Coefficient for Sunlight: $b_1 = 3.2$, SE = 1.2, t = 2.67, p = 0.01
  - Coefficient for Water: $b_2 = 0.8$, SE = 0.4, t = 2.00, p = 0.051
  - $R^2 = 0.52$
- (questions on next slide…)

Week 8 - stop and think

First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!

- In Model A, what does "3.8" mean in practical terms? ("For each additional hour of sunlight…")
- Why did the coefficient for Sunlight change from 3.8 to 3.2? And why did its standard error double (from 0.6 to 1.2)?
- Using the variance decomposition formula from slide 19, calculate:
  - How much variance is explained by Model A? (Hint: Use R^2)
  - How much additional variance is explained by adding Water?
  - Is this improvement substantial?
- Use the F-test approach (slide 35) to determine if Model B is significantly better than Model A.
  - You need: SSR for both models (use the relationship: $SSR = SST \times (1 - R^2)$)
  - Assume SST = 1000 for this example
  - Calculate F and interpret
- Advanced: Can you create a small R script to simulate this? Assume that Sunlight and Water have a correlation of 0.7 (get help from your AI…)